# Lesson 14: Raising the Stakes

Godot Engine

VIDEO GAME CREATION

**This week: later waves get genuinely harder - more enemies, moving faster - using the exact wave variable you've had since Lesson 11.**

**Open your own [YourName]Game project from Lesson 13 - don't start a new one.**

*Tip: no new assets this week - difficulty scaling reuses your exact existing enemy sprites and spawn logic.*

## Setup

1. Open your [YourName]Game project from Lesson 13.
2. No new variables needed for Level 1 - Level 2 will use enemy.tscn's existing speed variable (from Lesson 7).

## Level 1 - Minimal (MVP)

*Each wave spawns one more enemy than the last (5, 6, 7). Speed stays the same for now - that's Level 2.*

3. In spawn_wave(), change the fixed "for i in range(5)" loop into "for i in range(4 + wave)" (code below) - wave 1 spawns 5, wave 2 spawns 6, wave 3 spawns 7.
4. Wherever enemies_remaining gets reset to 5 for a new wave, change that to "enemies_remaining = 4 + wave" too - these two places must always agree.
5. Run the scene and play through a full round - wave 2 should visibly spawn one more enemy than wave 1, and wave 3 one more than that.

```
func spawn_wave():
    for i in range(4 + wave):
        var enemy = enemy_scene.instantiate()
        add_child(enemy)
        enemy.add_to_group("enemies")
        enemy.destroyed.connect(_on_enemy_destroyed)
        enemy.escaped.connect(_on_enemy_escaped)
        # ...existing per-enemy positioning...
    enemies_remaining = 4 + wave
```

## Level 2 - Complete (Alpha/Beta)

*Enemy speed ALSO increases each wave, on top of the growing count. Restarting correctly returns everything to wave 1's original values.*

- In spawn_wave(), right after instantiating each enemy, set its speed based on the wave: enemy.speed = 60.0 + (wave - 1) * 20.0.
- Confirm enemy.tscn's own speed variable (from Lesson 7) is still what its _process(delta) movement actually reads - no changes needed there.
- Restart mid-game and confirm both count and speed return correctly to wave 1's values.
- Play a full round and confirm wave 3 is noticeably busier and faster than wave 1, but still fair.

## Level 3 - Refined - Choose ONE Stretch Feature

*Pick exactly one of the five ideas below. These are intentionally open-ended - a good-faith attempt at the general idea is a complete Level 3 finish, not a fully polished feature.*

- Destructible shields/barriers: a few simple blocks placed between the player and the enemies that disappear (or shrink) after absorbing a hit or two - reuses your existing collision-detection and hit-reaction patterns from Lessons 4 and 13, just on a new kind of object.
- A power-up: an item that occasionally drops or spawns, which the player collects to temporarily change how shooting works (faster fire rate, multiple bullets at once, or a brief shield) - reuses your existing timer/cooldown pattern from Lesson 3.
- A bonus enemy: a rare, faster-moving, or oddly-patterned enemy worth extra points, spawned occasionally alongside a normal wave - reuses this week's own speed/count scaling ideas, just for one special case instead of the whole wave.
- Formation-marching movement: instead of each enemy independently drifting down (this course's approach since Lesson 4), have the whole wave move together as a group and occasionally step sideways - a genuinely different movement pattern, inspired by classic arcade shooters.
- A title screen and/or game-over screen: a proper opening screen (press a key to start) and/or a dedicated game-over screen with a final score, instead of just text overlaid on the gameplay - reuses your existing game_active flag and UI tools from Lessons 9-10. Godot's Tween node is a natural fit for a smooth title/game-over screen transition.

## If Something's Wrong

- Every wave still spawns exactly 5: confirm the loop actually uses "range(4 + wave)" and not a leftover range(5).
- WinText fires early or never: confirm enemies_remaining is reset using the SAME formula everywhere it's set, not 5 in one spot and 4 + wave in another.
- Speed doesn't change between waves: confirm enemy.speed is being set INSIDE spawn_wave() on each new instance, not left at enemy.tscn's default forever.

## Want More?

Student Resource List > Godot section, or docs.godotengine.org